

ReachNT

Remote repair triage that ranks by need, shares trips with H3 hexagons, and makes someone sign for who waits

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Town jobs always look cheaper, so remote tenants wait

5,058

remote public housing dwellings in 73 NT communities (DHLGCD)

5 vs 2

business days for an urgent repair, remote vs town, in the official NT standard (FS17)

8.5×

what an emergency repair costs in a very remote community vs a remote one (Nous 2017)

96%

of an emergency job's cost can be travel; batched planned work 11–37% (Nous 2017)

Santa Teresa tenants listed 600+ repairs in 2015. The courts held public housing must be at least safe, and the High Court allowed compensation for the distress of unrepaired homes (2023).

Left to “cheapest jobs first”, nobody decides who waits

62 days

Urgent repairs in remote communities: 90% done within this. Cut off in the wet: 79 days.

3 days

Urgent repairs in the five hub towns. Same schedule, same crews.

One synthetic year on real NT geography: 70 communities, 8,345 houses, 37,515 requests, 71 tradespeople. Cheapest-first costs \$558 a job, the lowest of any setting, which is why it looks reasonable on a cost report. 43% of remote urgent repairs pass even the official 5-day remote clock.

Rank by need. Share trips with hexagons. Price the gap

1

Read

Phrase rules + a text model turn a free-text report into a fault, trade and health practice. Unsure or dangerous: a person.

2

Most urgent first

Danger, health, who lives there and time waited set the order. Where someone lives never does.

3

Share trips

Nearby communities get one visit instead of two. The plan works around trades, wet-season roads and airstrips.

4

Show the cost of fairness

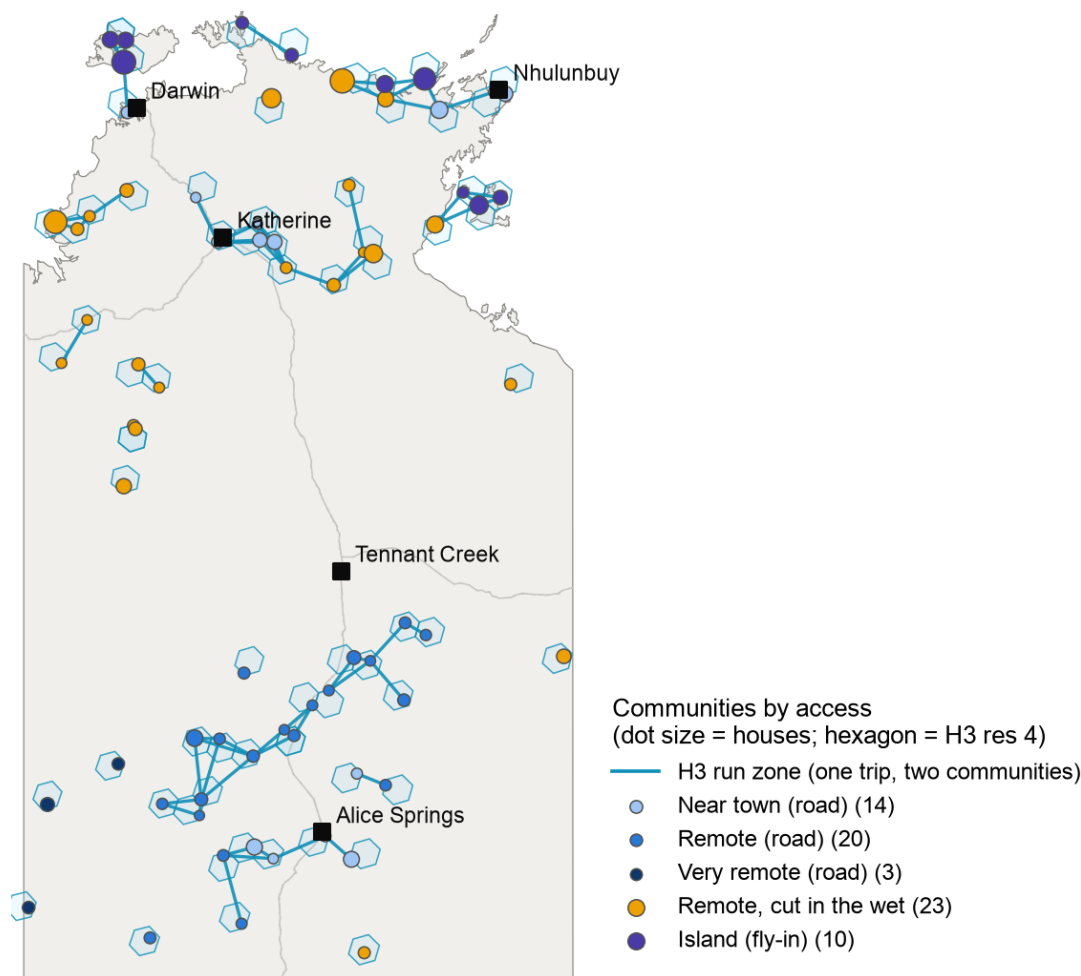
Each way of planning, side by side: dollars per repair against who waits.

5

Sign and explain

A named role signs a setting. Each week a job waits, the reason is logged and the tenant can see it.

H3: Uber's hexagon grid, used for remote repairs



Shared trips

Two communities whose hexagons are no more than 2 apart (about 90 km) get one visit: Katherine → A → B → home. 68 pairs, e.g. Ngukurr + Rittarangu.

Why hexagons

Six neighbours, all the same distance away, so “within k rings” is a fair circle. Square grids such as S2 have neighbours at two distances.

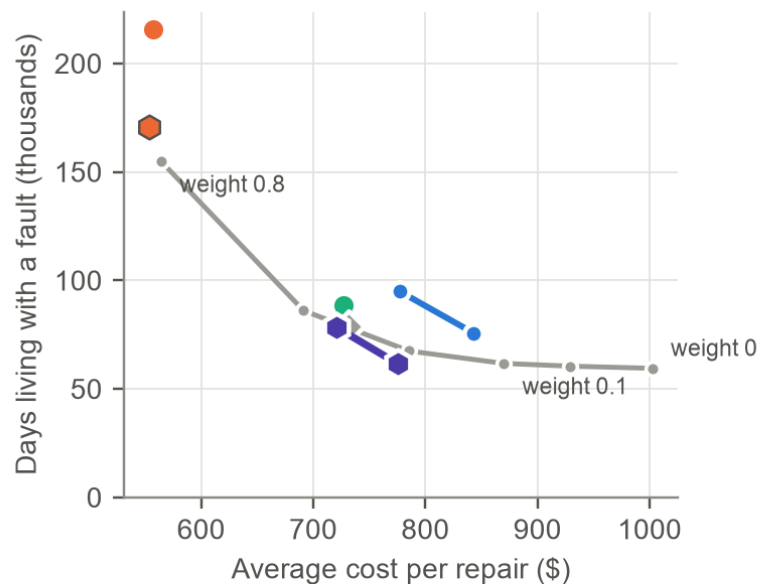
While you're out there

A tradesperson's phone lists open jobs within 3 resolution-5 rings of their run.

A house is a hexagon

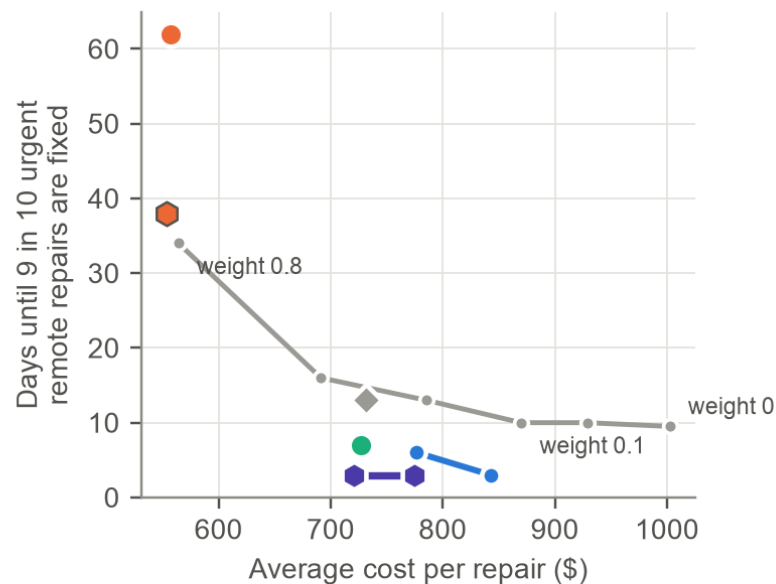
Records hold a 76 m resolution-10 cell, never an address. Public views use 36 km² cells, hidden below 5 jobs.

Fairness costs a little more, and shared trips cut the cost



● Urgent first, no deadline (cost weight 0 to 0.8)
● Urgent first, with a deadline
● ReachNT: urgent first, deadline, shared trips

● Cheapest first, with shared trips
● Cheapest jobs first



● Cheapest first, deadline for urgent jobs
● Urgent first, official remote deadlines

ReachNT vs cheapest-first

+\$218 per repair (+39%)
-71% days living with faults

9 in 10 urgent remote repairs fixed within
62 → 3 days

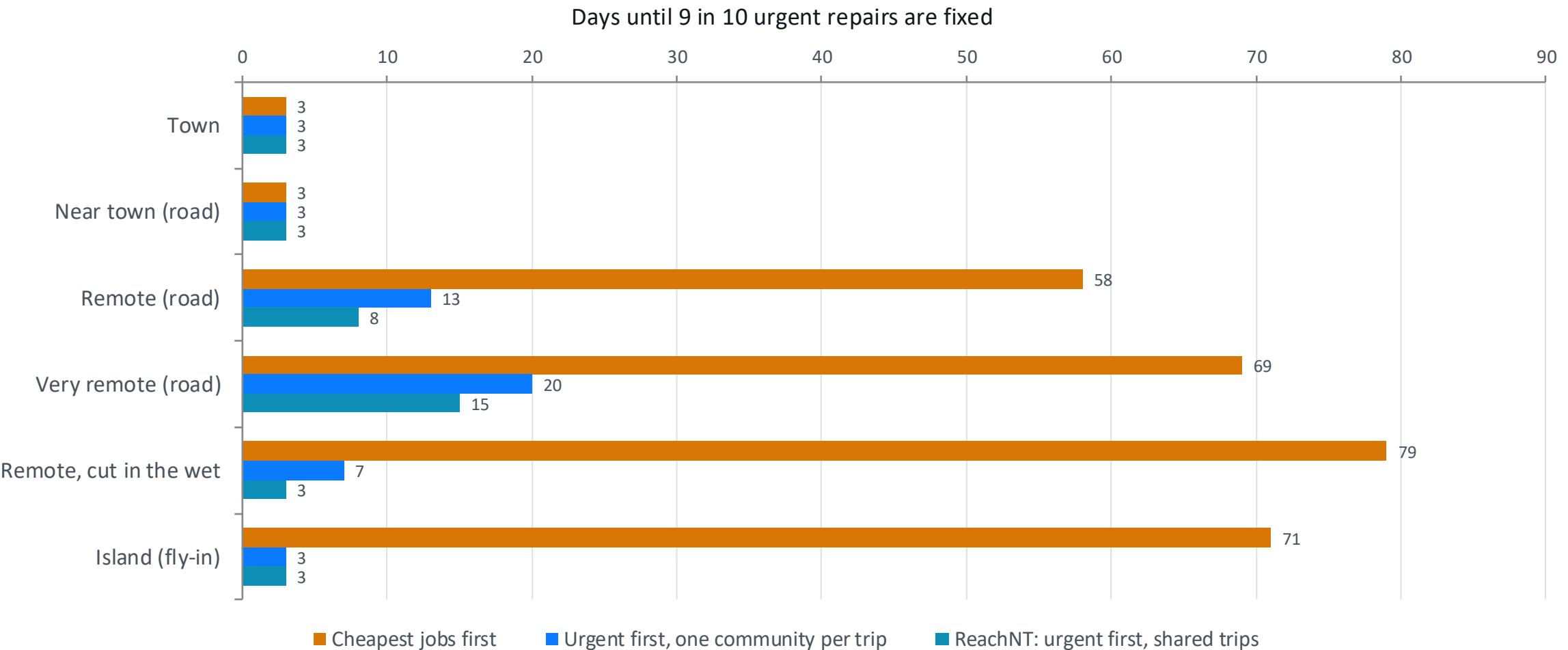
What shared trips added

\$843 → \$775 per repair

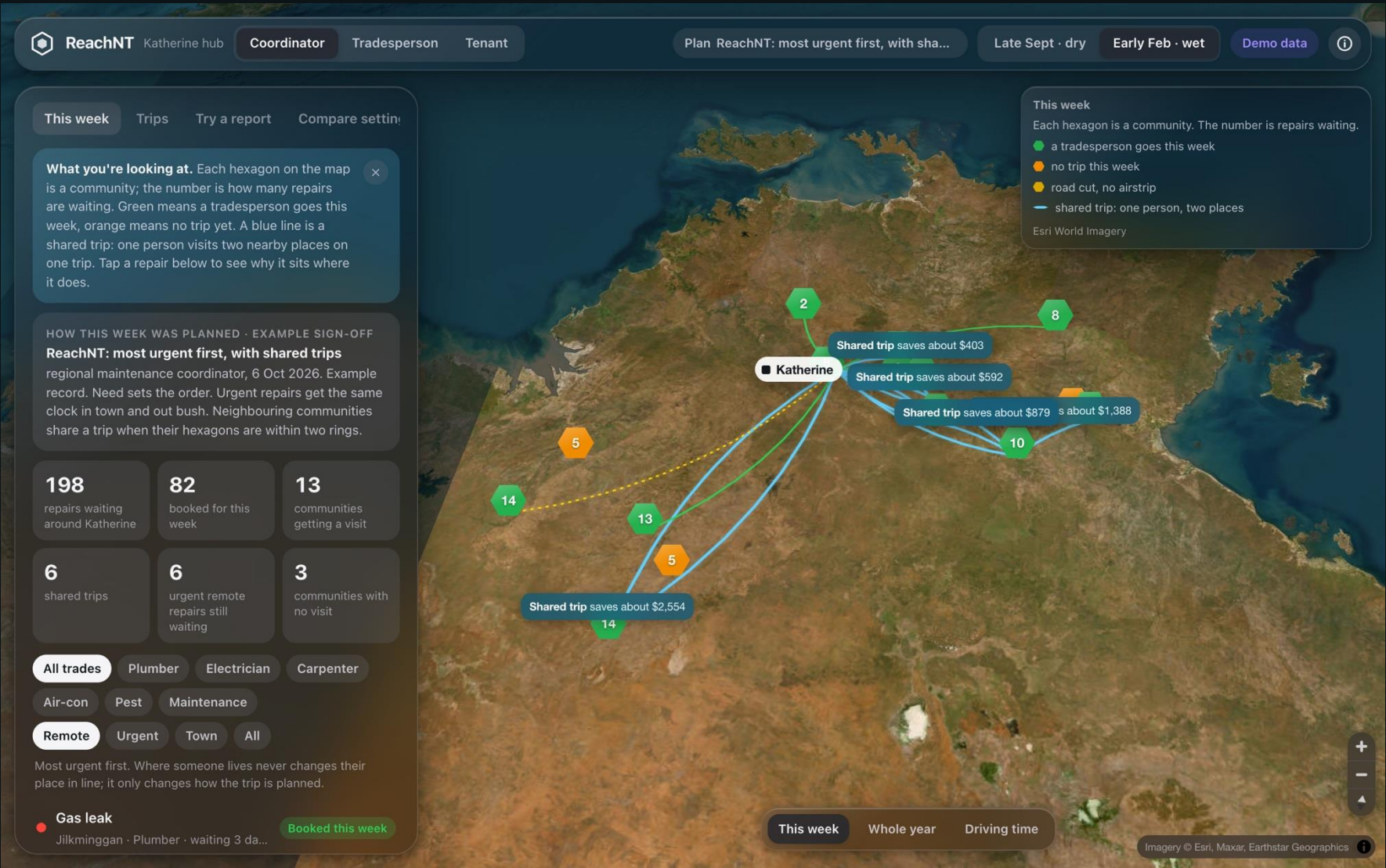
days living with faults 76k → 62k

Tighter budget: \$721 per repair, cheaper than
cheapest-first with a deadline (\$727).

Where the wait lands, by how hard the place is to reach



The coordinator sees the week on the real map



3D hexagon columns

Height is open repairs; colour is booked, waiting or cut off.

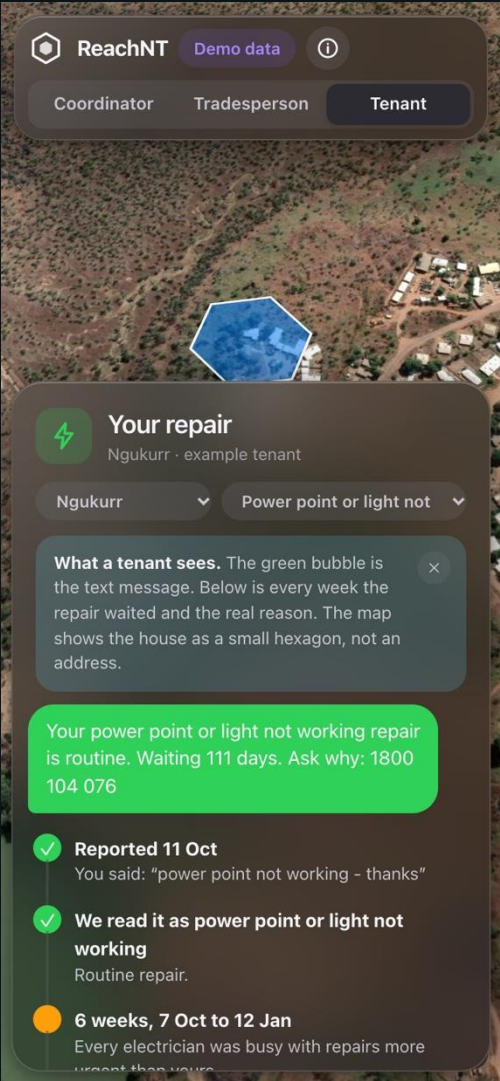
Shared trips

Blue lines join communities sharing a trip this week.

Satellite imagery

Esri online; Digital Earth Australia tiles bundled for offline use.

On the phone: a run sheet for trades, a tracker for tenants



Tradesperson

Only their own stops on the map, numbered in order. Each job shows what the tenant said, with Done or Couldn't do it. Save the run sheet as a PDF; updates made with no signal are sent later.

Tenant

A text message and a timeline built from the weekly reason log:
"For 21 days no electrician was sent to your community. It costs about \$2,000 to drive one there and back. That was a cost decision, not a judgement about your repair. Nobody signed off on it."

Names, phones and addresses stay locked away, on free tools

Vault (pii)

Name, phone, address, language.
Encrypted with pgcrypto. Read only by intake staff through a function that logs every read.

Work (ops)

Jobs, trips, reasons, the signed ledger. A house is an ID + a 76 m H3 hexagon.
Trades see their run; tenants see their own jobs.

Public

Waits by 36 km² hexagon and access band. Cells under 5 jobs hidden.

Stack: PostgreSQL + PostGIS + h3-pg + pgcrypto, all free and open source. Run it on NT Government servers for \$0 in licences, or pilot on Neon's Sydney region (supports all four; free tier 1 GB). A year of data is under 1 GB. Analytics in DuckDB with its H3 extension; the field app keeps data on the phone when there's no signal.

What it cannot do yet

AUC 0.78

for spotting a dangerous fault in wording the model never saw (1.00 on familiar wording). A person checks 40% of those reports, so 99% of dangerous ones are still caught.

Synthetic

repair requests: no public NT work-order data exists. Geography, roads, clocks and costs are real; comparisons are the finding, not the absolute numbers.

No co-design

yet. Real reports come in Kriol, Aboriginal English and other languages. Tenants, Aboriginal Housing NT, land councils and the AIS shape the words first.

We used satellite imagery of Country, not Aboriginal art. Any artwork would be commissioned and licensed through an Aboriginal art centre.

Make the remote allowance a signed decision

- DHLGCD: publish the triage rules; keep need and logistics separate; plan shared trips with H3; report urgent waits by access band; review the remote allowance each quarter with its cost.
- Contractors: log why each job waits; price charters, batching and shared trips into contracts; train local trades where waits are longest.
- Communities: co-design the tenant answer and fault words; hold the ledger to account.